Iteration D2: Connecting Products to Carts

We're looking at sessions because we need somewhere to keep our shopping cart. We'll cover sessions in more depth in *Rails Sessions*, but for now let's move on to implement the cart.

Let's keep things simple. A cart contains a set of products. Based on the Initial guess at application data diagram, combined with a brief chat with our customer, we can now generate the Rails models and populate the migrations to create the corresponding tables:

```
depot> bin/rails generate scaffold LineItem product:references cart:belongs_to
 ...
depot> bin/rails db:migrate
==  CreateLineItems: migrating ================================================
-- create_table(:line_items)
   -> 0.0013s
==  CreateLineItems: migrated (0.0014s) =======================================
```

The database now has a place to store the references among line items, carts, and products. If you look at the generated definition of the `LineItem` class, you can see the definitions of these relationships:

rails7/depot_f/app/models/line_item.rb

```
class LineItem < ApplicationRecord
  belongs_to :product
  belongs_to :cart
end
```

The `belongs_to` method defines an accessor method—in this case, `carts` and `products`—but more importantly it tells Rails that rows in `line_items` are the children of rows in `carts` and `products`. No line item can exist unless the corresponding cart and product rows exist. A great rule of thumb for where to put `belongs_to` declarations is this: if a table has any columns whose values consist of ID values for another table (this concept is known by database designers as *foreign keys*), the corresponding model should have a `belongs_to` for